Innovation document

New Menu Options

Option 7 - Update Attendee Details

Allows updating the details of an existing attendee. The following fields can be modified:

Name, Date of Birth (DOB), Gender, Company

Option 8 - Delete Attendee

Removes an attendee from the system after confirmation.

Actions performed:

- Deletes the attendee record from the MySQL database.

- Removes the corresponding node and relationships from the Neo4j database.

Option 9 - Show Connected Attendees in the Same Session

Displays all attendees who share at least one session with the selected attendee.

The output includes shared session details:

Session ID, Title, Date, Speaker, Room Name

Example:

```
Choice: 9

Show Connected Attendees in Same Session
------------------------------------------
Enter attendee ID: 101
------------------------------------------
Connected to Mia O'Brien, attendee ID: 107

Session ID | Title | Date | Speaker | Room Name
201 | Modern Data Pipelines | Dr. Niamh Burke | 2025-05-12 | Main Hall
------------------------------------------
Connected to Grace Power, attendee ID: 111

Session ID | Title | Date | Speaker | Room Name
202 | Scaling Neo4j for Recommendations | Prof. Alan Shaw | 2025-05-12 | Graph Lab
------------------------------------------
```

Note: Attendee ID 109 is not shown because they attend sessions 209 and 204, which are not shared with attendee ID 101.

Docker Compose and Database Seeding

The application can use Docker Compose to provision and initialize the required databases.

Docker Compose Configuration

The file resources/docker-compose.yml starts the following services:

- MySQL database

- Neo4j graph database

MySQL Initialization

The MySQL container automatically loads the schema and sample data from appdbproj.sql during startup.

This ensures the relational database is ready with predefined tables and example records.

Neo4j Initialization

Neo4j authentication is configured using the NEO4J_AUTH environment variable.

A health check ensures Neo4j is fully ready before dependent services run.

Neo4j Seeding Process

The neo4j-seed service executes resources/neo4j_seed.sh once Neo4j becomes healthy.

The script imports graph data from appdbprojNeo4j.json into Neo4j.

Reinitialization Behavior

If the Neo4j data volume is removed:

- The database is recreated on the next startup.

- The seeding process automatically runs again.

- All graph data is restored from the JSON seed file.

Summary

The system now supports:

- Updating attendee information

- Safely deleting attendees across relational and graph databases

- Discovering attendee relationships based on shared sessions

- Automated database provisioning and reseeding using Docker Compose